

# Shantanu Gorai

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## SUMMARY

Machine Learning Engineer and Full Stack Developer with applied experience building end-to-end AI systems — from data preprocessing pipelines and trained classification models to production-ready APIs and real-time dashboards. Specializes in anomaly detection, behavioral analytics, and network security intelligence, with a strong foundation in Python-based ML workflows and modern web technologies. Proven under pressure with national-level hackathon recognition and industry training at Tata Steel.

## EXPERIENCE

### Vocational Trainee — Image Recognition using Machine Learning

Apr 2026 – May 2026

Tata Steel Ltd., Learning & Development Department

Jamshedpur, Jharkhand

- Developed an image recognition system using supervised ML techniques within Tata Steel's industrial training environment.
- Designed and evaluated classification pipelines on real-world image datasets, applying feature extraction and model performance tuning.

## TECHNICAL SKILLS

|                            |                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>ML Frameworks:</b>      | Scikit-learn, NumPy, Pandas, Scapy (network data processing)                                                  |
| <b>Core Domains:</b>       | Anomaly Detection, Network IDS, Behavioral Analytics, Time-Series Modeling, Spatial Mapping (SLAM)            |
| <b>ML Methods:</b>         | Feature Engineering & Selection, Dimensionality Reduction, Classification & Regression, Behavioral Clustering |
| <b>Backend &amp; APIs:</b> | Python, Flask, Node.js, Express.js, REST APIs (Model Deployment & Serving), JWT Authentication                |
| <b>Frontend:</b>           | React.js, Next.js, TypeScript, Tailwind CSS                                                                   |
| <b>Security:</b>           | Network Packet Inspection, Telemetry Analysis, Cryptography (SHA-256 / Blowfish), Secure Coding Practices     |
| <b>Tools:</b>              | Git, GitHub, Vercel, Render, PostgreSQL, MongoDB, Linux                                                       |

## PROJECTS

### Real-Time Network Intrusion Detection System (IDS) | Python, Flask, Scikit-learn, Scapy, React.js, WebSockets 2025

- Trained a supervised classification model on network telemetry features to distinguish benign vs. malicious traffic with real-time inference.
- Built a live preprocessing pipeline using Scapy to parse raw packet attributes on the fly as direct model input.
- Served ML predictions via Flask REST API with WebSocket streaming to a React.js + Tailwind CSS threat dashboard at sub-second latency.

### AI Student Productivity & Behavioral Analytics Dashboard | Python, Scikit-learn, React.js, Next.js, PostgreSQL 2025

- Engineered a behavioral analytics engine processing time-series data — timetables, sleep patterns, mood logs — to predict student stress risk.
- Applied clustering and time-series modeling to surface productivity insights and personalized recommendations via a Next.js frontend.
- Built full-stack application with Python analytics backend, JWT authentication, and PostgreSQL persistence.

### Geographic Velocity Anomaly Engine (Impossible Travel Detector) | Python, Flask, IP Geolocation API, Risk Scoring 2025

- Designed an algorithmic risk-scoring engine detecting compromised credentials by computing geographic velocity between consecutive login events.
- Integrated live IP Geolocation APIs to dynamically flag login sequences that violate physical travel constraints, triggering automated security alerts.

## EDUCATION

### Bachelor of Engineering – Computer Science

2023 – 2027

ACS College of Engineering, Bengaluru (VTU Affiliated)

Bengaluru, Karnataka

## ACHIEVEMENTS

- Runner-Up, CODE CARNAGE – National Level Hackathon** (Apr 2025): Built *VocalLock*, an AI Voice Authentication & Spoof Detection System.
- Semi-Finalist, CODE BREAKER Challenge** (May 2025): Demonstrated strong ML problem-solving and full-stack software development skills.
- Top-5 Finalist, AI-VERSE 2.0** (Oct 2025): Designed and presented an innovative AI-powered solution among nationally competing teams.